Bomb Diffusal Activity

This project will combine a Raspberry Pi with some tools and prompt students in teams of two to diffuse the bomb. The program will choose 6 of the possible puzzles in a random order and show a 30 minute timer at the top of the screen at all times. If they get through all 6, they win!

- First, create the two manuals that contain the items/placeholders for items that require more planning.
- Second, create a parts list for the rpi “bomb”
- Third, create a program for the rpi.

Materials List

Player 1	Player 2
- Screen - Mouse - Keyboard - Decoder Cards - morse code translator - simon “riddles”	- 2 RGB LEDs - 4 colored buttons (R, Y, G, B) - Rotary Encoder - Manual - Maze - Map, ruler and protractor - List of symbol traits - List of symbols and numbers

Puzzle Design

Puzzle 1: Color Mixing

- A random hexadecimal value is shown on one RGB led. The team must use R, G, and B sliders to match the color with a second LED

Player 1	Player 2
- Is shown 3 sliders (one for red, green and blue but they are unlabeled)	- Can see both the target and team’s LED

Puzzle 2: Morse Code

- A sequence of Morse code is played on an LED. Must correctly type in the translation to the keyboard

Player 1	Player 2
- Has a card that translate morse code - Is shown blanks on a screen. Must type in the correct solution	- Can see one LED blinking a patten

Puzzle 3: Simon with a Twist

- A sequence of colors is shown on the LED, following the translation rules from the decoder cards, the team must click the correct sequence on the buttons

Player 1	Player 2
- Has a card that translates colors. For example, green => red, blue => 2 yellows, etc.	- Can see one LED blinking a color pattern - Can type in a sequence using the buttons

Puzzle 4: Invisible Maze

- The screen shows a red dot on the left side, must navigate it use arrow keys to the labeled goal

Player 1	Player 2
- Uses the arrow keys to move the dot through a maze that they cant see	- Manual contains the maze

Puzzle 5: Navigation

- The screen shows a list of directions, “starting at the start of the trail, go three miles after a 30 degree left turn etc. The screen prompts the user, where did I end up?

Player 1	Player 2
- Relays instructions on the screen then inputs the final location.	- Manual contains the map, a ruler and a protractor.

Puzzle 6: Turn on the LED

- The screen shows a circuit diagram with AND, OR, NOT, and XOR gates. Students toggle input switches by clicking them until the output lamp lights.

Player 1	Player 2
- Clicks the LEDs until the output lights	- Manual contains a circuit diagram of which are and or and not

Puzzle 7: Rhythm Lock

- A rhythm is played over the speaker. Must use the buttons to play it back

Player 1	Player 2
- Can see the note names appearing as they are pressed	- Presses buttons to replay the sound

Puzzle 8: Symbols Lock

- The screen shows 5 symbols, and 4 of them share a hidden trait: same number of strokes, all curved, all symmetrical, or all closed shapes. Students click the one that doesn’t belong

Player 1	Player 2

- Can see the symbols - Click the incorrect section	- The manual lists the traits but not which one applies

Puzzle 9: Glyph Addition

- The manual gives each symbol a number value. The screen shows an equation using symbols, such as ◇ + ⌘ − ✶, and students type the numeric answer.

Player 1	Player 2
- Can see the equation	- The manual lists a number value for each symbol.

Puzzle 10: Rotation Lock

- The screen shows two symbols, one rotated with the rotary encoder. Must spin the encoder so they match for three rounds within a time limit

Player 1	Player 2
- Can see the symbols	- Spins the rotary encoder